

PipeFit Pro — finding the common measurements

Every screen is reached from the home list. Type what you measured; read what you cut.

1. Offset around an obstruction — 45°

The one you do most.

Simple offset → **Offset** = how far you have to move over → tap preset **45°**.

Read **PIPE CUT**. That is the piece between the two fittings, gap already off.

12" offset, 2" pipe, 45° → travel 16.97, setback 1.24 each end, **cut 14.30**
(14.49 before the two 3/32" weld gaps come off)

Run fills itself in: at 45° it equals the offset.

2. Offset at any other angle

Same screen. Presets are **11.25° · 22.5° · 30° · 45° · 60° · 90°**, or type any angle into **Fitting angle**.

3. Square jog — 90°

Preset **90°**. The pipe goes straight across and advances nothing.

Run = 0. Travel = the offset.

24" offset, 2" pipe → **cut 17.81** (24 – 3 – 3 – two 3/32" gaps)

4. When the run is fixed, not the angle

You know how much room you have along the line, and the angle has to fall out of it.

Simple offset → **Offset** → tap **Set run** → type the run.

The angle is solved for you and shown as **Cut angle**. A run of **0** gives you the square jog.

5. Rolling offset — moves two ways at once

Rolling offset → **Rise / set** (up or down) → **Roll** (left or right) → tap an **Elbow angle** preset.

- **True offset** — the real diagonal distance, straight through
- **Cut angle** — what you set the fittings to
- **PIPE CUT** — the piece between them

12" up, 9" over → true offset **15"**, roll angle **36.87°**

To fix the run instead of the angle, tap **Run** and type it.

6. Cut length between two fittings

You have a centre-to-centre dimension off a drawing and a fitting on each end.

Cut length → **C2C length** → pick **Joint** (screwed / welded / flanged / soldered) → pick **End A** and **End B**.

Read **PIPE CUT**.

24" C2C, 2" pipe, two long radius 90s → cut 18"

Flanged adds a **Class** row — 150, 300, 600 and up. Anything not on the list: choose **Custom** and type the takeout.

7. Screwed pipe — how far it makes up

Thread engagement → **Size** → **C2C length**.

Gives makeup, takeout and tap drill. Override the fitting with **Fitting C-to-face** if yours is not standard.

8. Bending your own pipe

Pipe bend → radius and angle → setback, arc length and gain, with the bend marks.

Saddle bend for three and four point saddles. **Miter bend** for a segmented elbow — it checks the B31.3 22.5° limit for you.

9. Looking a figure up

Handbook → search. Every table carries its page number.

10. Checking the app in your head

Multiply the offset by these. The app agrees to four decimal places — it is tested against them.

FITTING ANGLE	TRAVEL	RUN	SHRINK
11.25°	× 5.126	× 5.027	× 0.098
22.5°	× 2.613	× 2.414	× 0.199
30°	× 2.000	× 1.732	× 0.268
45°	× 1.414	× 1.000	× 0.414
60°	× 1.155	× 0.577	× 0.577
90°	× 1.000	× 0	× 1.000

Travel = centre to centre of the two fittings. **Run** = how far the offset advances along the line. **Shrink** = travel – run, what the offset costs you.

11. The one trap — a bend is not a bought fitting

They are the same at 90° and different everywhere else.

2" LONG RADIUS 45	TAKES OUT
Bent pipe	1.2426"
Bought elbow	1-3/8"

A run of bought fittings worked to the bend formula cuts **every piece an eighth long**.

The **Cut length** screen offers both, labelled. The **Simple offset**, **Rolling offset** and **Pipe bend** screens work the bend, which is what a bender does.

Settings worth setting once

- **Weld gap** — default 3/32". Comes off welded and flanged ends only; a screwed or soldered joint pulls up tight and never has it deducted.
- **Fractions** — off, 1/8, 1/16, 1/32, 1/64. The decimal under it is always the exact figure. Cut to the decimal when tolerance is tight.
- **Default size** — the pipe every screen opens on.

The 3D spool

3D spool → add legs, each with a length, a bend angle off the last one and a roll.

- **Drag** to turn it. The rotation is the sweep where nothing goes edge on — about 260°, with the dead views taken out rather than passed through.
- **Hold a leg** in the picture and drag along it to stretch or shorten it.
- **NE / NW / SW / SE** jump to the four isometric corners.